

Selected Presentations

Rachel N. Slaybaugh

August 16, 2021

“Optimal Operations of Advanced Reactors: How Do We Get There?” Current Issues in Computational Methods – Roundtable, American Nuclear Society Annual Meeting. Online. 14 June 2021. (invited)

Guest Lecture “Nuclear Energy: Why?” for Energy 790-1 MSE Energy Class at Georgia Tech. Online. 1 Apr. 2021. (invited)

Panelist for “Serving the Nation: Engineering as Leaders in Government.” Spring Dean’s Society Event. Online 24 Mar. 2021. (invited)

“How Can Nuclear Support a Clean Energy Future?” American Physical Society Meeting. Online. 17 Mar. 2021. (invited)

“Nuclear in the 2020s: considerations for the decade ahead.” **Keynote Speaker**, ARPA-E Annual Fission Meeting. Online. 23 Feb. 2021. (invited)

Guest Lecture “Nuclear Energy (where we are, where we are going, and why)” for STIA 402: Clean Energy Innovation, Georgetown University School of Foreign Service. Online. 20 Oct. 2020. (invited)

Guest Lecture “Nuclear Energy: Why?” for Energy 790-1 Emerging Energy Technologies Class at Duke University. Online. 15 Sept. 2020. (invited)

Emcee for “Good Energy Collective Meet and Great.” Online. 11 Aug. 2020.

Spoke in various sessions at 2020 Nuclear Innovation Bootcamp. Online. 6-7 Aug. 2020.

Panelist for “Flexible Nuclear for Future Energy Systems Webinar.” Lucid Catalyst and ClearPath. 5 Aug. 2020. (invited)

Panelist for “Beyond Now: Microreactors and Near Term Demonstrations - Perspectives from the Department of Defense and Department of Energy.” Panelist. U. S. WIN Virtual Conference. 28 July 2020. (invited)

Panelist for “Advanced Nuclear Tech in a Decarbonized Energy Future: Feasible or Fantastical?” Duke Energy Initiative lecture series. Online. 14 July 2020. (invited)

“How Can Nuclear Be Part of a Clean Energy Future?” Breakthrough Generation Bootcamp. Online. 18 June 2020. (invited)

Panelist for “Challenges and Opportunities in Thermal Hydraulics of High Temperature Gas Cooled Reactors.” American Nuclear Society Annual Meeting. Online. 8 June 2020. (invited)

Panelist for “Building Your Leadership Competency.” American Nuclear Society Annual Meeting. Online. 8 June 2020. (invited)

“Nuclear Engineering Research to Make a Difference.” UC Berkeley NE Dept. Graduate Colloquium. Berkeley, CA. 9 March 2020.

“How Can Nuclear Be Part of a Clean Energy Future?” G20 Workshop on Circular Carbon Economy. Riyadh, KSA. 5 March 2020. (invited)

“Special Update: ARPA-E Nuclear Program Update.” Advanced Reactors Summit VII. Knoxville, TN. 11 Feb 2020. (invited)

“The Opportunity of Next-Gen Nuclear Energy.” Citizen’s Climate Lobby, Fairfax Meeting. Annandale, VA. 11 Feb 2020. (invited)

Moderator for “Deploying Capital for a Clean Energy Future.” BERC Energy Summit 2020. Berkeley, CA. 7 Feb 2020. (invited)

“How Can Nuclear Energy Be Part of a Clean Energy Future?” Climate Tech Convening 2020. San Francisco, CA. 30 Jan 2020. (invited)

Panelist for “Time to be Inclusive and Pragmatic Before Its Too Late: Advanced nuclear technologies must be a significant part of the solution set.” Cleantech Forum. San Francisco, CA. 28 Jan 2020. (invited)

“What is ARPA-E Doing in Nuclear Energy?” Argonne National Laboratory Colloquium. Lemont, IL. 20 November 2019. (invited)

“What is ARPA-E Doing in Nuclear Energy?” American Nuclear Society Winter Meeting MEITNER Special Session. Washington, DC. 18 November 2019.

Guest Lecture “Nuclear Energy” for STIA 402: Clean Energy Innovation, Georgetown University School of Foreign Service. 23 Oct. 2019. (invited)

Panelist for “The role of innovation in the civilian and military nuclear energy sectors.” The civilian-military nexus and the role of innovation in nuclear energy, Atlantic Council. Washington, DC. 15 Oct 2019. (invited)

“Next Generation Nuclear.” Ecomodernism 2019. Middleburg, VA. 23 September 2019. (invited)

“Nuclear and ARPA-E: Activities and Opportunities.” NUC Workshop: Innovations in Advanced Reactor Design, Analysis, and Licensing. Raleigh, NC (given remotely). 18 September 2019. (invited)

“Nuclear Energy Programs and Research at Advanced Research Projects Agency-Energy (ARPA-E).” GABI nuclear energy roundtable. Washington, DC. 12 September 2019. (invited)

“Optimal Operations of Advanced Reactors: If You Build It, Can you Run It?” ARPA-E Program Pitch. Washington, DC. 23 July 2019.

“What if Nuclear Plants Were 10x Cheaper to Operate?” ARPA-E Energy Summit. Denver, CO. 10 July 2019.

Panelist for “Future Scientists and Engineers for Nuclear Energy.” Third Symposium on US-Japan Nuclear Energy Research Cooperation. Washington, DC. 26 June 2019. (invited)

Panelist for “Lessons from Solar PV.” Breakthrough Dialogue 2019 Whole Earth Discipline. Sausalito, CA. 21 June 2019. (invited)

“Computing in Nuclear Engineering: Moving Beyond Today’s Limitations.” UW Madison Engineering Physics. Madison, WI. 28 May 2019.

“Computing in Nuclear Engineering: Moving Beyond Today’s Limitations.” NC State Nuclear Engineering Colloquium. Raleigh, NC. 23 May 2019.

“Computing in Nuclear Engineering: Moving Beyond Today’s Limitations.” GA Tech Nuclear & Radiological Engineering Colloquium. Atlanta, GA. 10 May 2019.

“Computing in Nuclear Engineering: Moving Beyond Today’s Limitations.” MIT Nuclear Science & Engineering Colloquium. Cambridge, MA. 29 April 2019.

“Computing in Nuclear Engineering: Moving Beyond Today’s Limitations.” TAMU Nuclear Engineering Colloquium. College Station, TX. 16 April 2019.

“Creating the Future of Nuclear Energy.” U Michigan Nuclear Engineering and Radiological Sciences Colloquium. Ann Arbor, MI. 18 Jan 2019. (invited)

“Creating the Future of Nuclear Energy.” CITRIS Research Exchange Seminar Series. Berkeley, CA. 28 Nov. 2018. (invited)

“Creating the Future of Nuclear Energy.” Berkeley Distinguished Lectures in Data Science. Berkeley, CA. 27 Nov 2018. (invited)

Panelist for “Ethics in Nuclear Engineering and Design.” American Nuclear Society. Orlando, FL. 13 Nov. 2018. (invited)

“U.S. Nuclear Innovation System.” Commercialisation of Small Nuclear in the UK. Coventry, UK. 6 Nov. 2018. (invited)

“MEITNER: The Newest ARPA-E Program.” NASA iTech. New York, NY. 14 June 2018. (invited)

“Deep decarbonization: financing the rapid deployment of innovations after R&D.” 2050 Today Panel. San Francisco, CA. 15 June 2018. (invited)

“Discussion of Future Technology Development.” Nuclear R&D Summit Panel. Washington, DC. 7 Mar. 2018. (invited)

“Advanced Solver for Radiation Transport.” North Carolina State University Dept. Graduate Colloquium. Raleigh, NC. 1 Feb. 2018. (invited)

“Advanced Solver for Radiation Transport.” Colorado School of Mines NSE Program Graduate Colloquium. Golden, CO 17 Jan. 2018. (invited)

“Advanced Solvers for Radiation Transport.” University of Illinois Urbana Champaign NPPE Dept. Graduate Colloquium. Champaign, IL. 5 Dec. 2017. (invited)

Moderator for “Commercialization of Nuclear Fusion Summit” Moderator, Nuclear Fusion: Reacting To Commercialization by VLAB. Palo Alto, CA. 14 Nov. 2017. (invited)

“Nuclear Energy: What Can Analytics Do for Economics and Safety?” **Keynote Speaker**, Industrial Machine Learning Workshop 17. San Francisco, CA. 24 Oct. 2017. (invited)

Panelist for “Alternative energy, nuclear and CCS: What are the prospects?” Low-Emissions Solution Conference at the 5th International Conference on Sustainable Development. New York, NY. 21 Sept. 2017. (invited)

Panelist for “The Case for Nuclear: Communications and Analyzing Risks and Benefits” Aspen Institute Future of Nuclear Energy Roundtable. Aspen, CO. 12 Sept. 2017. (invited)

“Advanced Solvers and Innovation for Penetrating Radiation” **Keynote Speaker**, SPIE Penetrating Radiation Technical Event. San Diego, CA. 8 Aug. 2017. (invited)

“Advancing Solvers and Challenges for Nuclear Innovative.” Lawrence Berkeley National Laboratory, IBT Seminar. Berkeley, CA. 12 May 2017. (invited)

“Challenges & Opportunities in Developing an Innovation Culture in Nuclear.” Second Annual Third Way Advanced Nuclear Summit Panel. Washington, DC. 21 Feb. 2017. (invited)

“Advancing Reactors with Advanced Solvers and Innovative Programs.” Electric Power Research Institute. Research Meeting. Palo Alto, CA. 13 Feb. 2017. (invited)

“Exploring the Advanced Nuclear Ecosystem.” SxSW Eco Panel. Austin, TX. 12 Oct. 2016. (invited)

“Nuclear and Innovating the Clean Energy Landscape.” French Nuclear Energy Society Young Generation Workshop. Video Meeting. 28 Sept. 2016. (invited)

“Nuclear and Innovating the Clean Energy Landscape.” NorCal ANS Meeting. Berkeley, CA. 21 Sept. 2016. (invited)

“There Is So Much Room for Leadership.” 2016 Nuclear Innovation Bootcamp Closing Speaker. Berkeley, CA. 12 Aug. 2016.
<https://www.youtube.com/watch?v=imvnBYa9Sb0> (starting at 12 min)

“Nuclear Innovation: Concept to Execution.” PHYSOR 2016. Sun Valley, ID. 2 May 2016. (invited)

“The Opportunity of Global Nuclear Innovation.” UC Berkeley NE Dept. Graduate Colloquium. Berkeley, CA. DATE??? 2016. (invited)

“The Opportunity of Global Nuclear Innovation.” 49th Japan Atomic Industrial Forum (JAIF) Annual Conference. Tokyo, Japan. 12 April 2016. (invited)

“Exascaling Nuclear Innovation.” **Keynote speaker**, Conference on Data Analysis 2016. Santa Fe, NM. 4 Mar. 2016. (invited)

“Exascaling Nuclear Innovation.” BIDS Distinguished Lecture Series. 19 Feb. 2016. (invited)

“Lets Get Real: When can we expect commercial advanced reactors?.” The Advanced Nuclear Summit and Showcase Panel. Washington, DC. 27 Jan. 2016. (invited)

“Hybrid Transport Methods for Shielding Challenges.” Idaho National Laboratory seminar. Idaho Falls, ID. 20 Nov. 2015. (invited)

K.L. Rowland, R.N. Slaybaugh. “Developments in the GPU-accelerated WARP Monte Carlo Neutron Transport Code.” Workshop sur l’utilisation des accélérateurs (GPUs, MICs) pour les simulations Monte-Carlo. École Polytechnique, Saclay, France. 10 July 2015.

R.N. Slaybaugh, S.C. Wilson. “Solving Shielding Challenges: Self-Shielding and Strong Anisotropies.” Oak Ridge National Laboratory Summer Intern Series. Oak Ridge, TN. 11 Jun. 2015. (invited)

K.L. Rowland, R.M. Bergmann, R.N. Slaybaugh, J.L. Vujic. “Implementing delta-tracking in a GPU-accelerated Monte Carlo neutron transport code.” 2nd Frontiers in Computational Physics Conference: Energy Sciences. Zurich, Switzerland. 2-5 Jun. 2015.

R.N. Slaybaugh. “Solving Shielding Challenges: Self-Shielding and Strong Anisotropies.” University of Florida NE Dept. Graduate Colloquium. Gainesville, FL. 2 Apr. 2015. (invited)

R.N. Slaybaugh, T.M. Evans, and S.W. Mosher. “Improved Hybrid Modeling of Used Fuel Storage Facilities.” DOE-NE MPACT meeting. Oak Ridge, TN. 24-26 Mar. 2015. (invited)

R.N. Slaybaugh. “Academia Panel.” WiSE O.N.E.: Theory Into Practice conference . Berkeley, CA. 14 Mar. 2015. (invited)

“Hybrid Methods for Shielding Challenges: Self-Shielding and Strong Anisotropies.” Colorado School of Mines NSE Program Graduate Colloquium. Golden, Colorado. 7 Jan. 2015. (invited)

“Computational Science in the ‘Real World’.” Computer Science I and II, Baldwin Upper School. Bryn Mawr, PA. 10 and 11 Dec. 2014. (invited)

“PyNE and Nuclear Data: A Next Generation Tool.” Sather Next Generation Nuclear Science Meeting. Lawrence Berkeley National Laboratory. 9 Dec. 2014. (invited)

“Advanced Approaches to High-Performance Computing in Nuclear: Applications to Non-Proliferation.” MIT Delegation Visit to BNRC. Berkeley, CA. 8 Dec. 2014.

“Computational Methods and Software Development in Nuclear Engineering Research.” Tea at Berkeley Institute for Data Science. Berkeley, CA. 4 Dec. 2014. (invited)

“Transitions Panel” and “Summary Roundtable.” Expanding Potential: A workshop on navigating the hurdles faced by women in STEM. Berkeley, CA. 15 Nov. 2014. (invited)

- “The PyNE Software Library: A Framework for ENSDF?” Nuclear Data Week Meeting. Brookhaven National Laboratory. 6 Nov. 2014.
- “The PyNE Software Library: Why and How?” ANS Northern California Local Section Meeting. Alfred’s Steakhouse, San Francisco, CA. 22 Oct. 2014. (invited)
- “What’s New in Academia Panel.” U.S. Women In Nuclear Annual Meeting. Marriott Copley Place, Boston, MA. 29 July 2014. (invited)
- “The Resonance Factor Method: Accelerating Monte Carlo in the Presence of Space and Energy Self-Shielding.” CEA-Saclay Colloquium. Saclay, France. 26 June 2014.
- “The Resonance Factor Method: Accelerating Monte Carlo in the Presence of Space and Energy Self-Shielding.” Reactor Physics Methods and Analysis Group Seminar. Areva Headquarters, Paris, France. 27 June 2014.
- R.N. Slaybaugh, T.M. Evans, P.P.H. Wilson, S.C. Wilson. “Radiation Transport: Computational Methods and Real-World Use.” UC Berkeley NE Dept. Graduate Colloquium. Berkeley, CA. 29 Oct. 2012. (invited)
- R.N. Slaybaugh, T.M. Evans, P.P.H. Wilson, S.C. Wilson. “Radiation Transport: Computational Methods and Real-World Use.” NC State Univ. NE Dept. Graduate Colloquium. Raleigh, NC. 8 Nov. 2012. (invited)
- “Acceleration Methods for Massively Parallel Deterministic Transport.” KAPL Employment Meeting. Niskayuna, NY. 30 Aug. 2011. (invited)
- R. Slaybaugh, M. Arbidze, S. Lamichhane, D. O’Connor. “An Evaluation of European Union Energy Policies.” UW–Madison Center for World Affairs and the Global Economy Seminar. Madison, WI. 11 May 2011.
- “Krylov Methods and JFNK.” UW—Madison Radiation Hydrodynamics Meeting. Madison, WI. 16 Dec. 2010. (invited)
- R.N. Slaybaugh, T.M. Evans, G.G. Davidson. “Parallel Algorithms for Fixed-Source and Eigenvalue Problems.” 2010 SIAM Annual Meeting. Pittsburgh, PA. 12-16 July 2010.
- “Variance Reduction in MC21 using Forward Adjoint Variance Reduction (FAVRE).” Naval Reactors—Shielding Video-conference. Pittsburgh, PA. Aug. 2010.
- “MC21—Jaguar Coupling for Variance Reduction.” KAPL Physics Forum. Niskayuna, NY. July 2009.